THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 8 — SCIENCE · MEDIUM LEVEL MCQ TEST SET 2

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Medium level: multi-step questions.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. A sudden increase in the population of rabbits in a grassland may occur if: 1

(a) grass is removed **(b)** rains are heavy
(c) owls are added **(d)** foxes are removed

2. Venus has a surface temperature of about 460 degrees Celsius mainly due to: 1

(a) a thick CO2 atmosphere **(b)** closeness to the Moon
(c) its oceans **(d)** its magnetic field

3. For a study with the dependent variable 'height of a plant', the most suitable unit is: 1

(a) kg **(b)** s **(c)** degree C **(d)** cm

4. A student gets a 400x total magnification using a 40x objective lens. The eyepiece lens is: 1

(a) 100x **(b)** 20x **(c)** 10x **(d)** 5x

5. Vaccines are more effective when given: 1

(a) after the symptoms appear **(b)** only to adults
(c) before exposure to disease **(d)** only after recovery

6. A scrap yard crane drops the iron when: 1

(a) the wire is long **(b)** the current is switched off
(c) the current is on **(d)** the iron is heavy

7. A charged balloon attracts a neutral wall because: 1

(a) it induces opposite charge near it **(b)** of gravity only
(c) the wall is a magnet **(d)** the balloon is heavy

8. A sealed bag of chips puffs up on a hilltop because: 1

(a) outside pressure is higher **(b)** outside pressure is lower than inside
(c) chips expand **(d)** air inside cools

9. A gas can be turned into a liquid if it is cooled below its: 1

(a) room temperature **(b)** freezing point only
(c) melting point only **(d)** boiling point (condensation point)

10. Which of the following is NOT an alloy? 1

(a) copper **(b)** brass **(c)** bronze **(d)** steel

11. Which of the following will show the Tyndall effect? 1

(a) milk **(b)** sugar solution **(c)** salt solution **(d)** vinegar

12. A mirror is needed for the side of a car to show a wide area behind the driver. The best choice is a: 1

(a) plane mirror **(b)** convex mirror
(c) cylindrical glass block **(d)** concave mirror

13. The difference between the solar year (365 days) and the lunar year (354 days) is about: 1

(a) 30 days **(b)** 11 days **(c)** 1 day **(d)** 5 days

14. Burning of coal increases carbon dioxide and the effect on the atmosphere is: 1

(a) global warming **(b)** more oxygen **(c)** cooling **(d)** no effect

15. The inner core is solid even though it is very hot because of: 1

(a) extremely high pressure **(b)** low temperature
(c) water **(d)** no gravity

16. The time for an object to fall decreases when it is dropped from a lower height. The variables that are related are: 1

(a) mass of earth and shape **(b)** colour and time
(c) shape and colour **(d)** height and time

17. A magnification of 100x means an object of 0.1 mm appears as: 1

(a) 0.01 mm **(b)** 100 mm **(c)** 10 mm **(d)** 1 mm

18. The skin protects us because it is: 1

(a) the organ that makes antibodies **(b)** a layer that kills every germ
(c) a physical barrier to germs **(d)** the organ that makes white blood cells

19. A student tests materials in a circuit: bulb glows with a coin, key and iron nail; not with eraser, wood and glass. The materials that conduct are: 1

(a) eraser, wood and glass **(b)** all of them
(c) coin, key and iron nail **(d)** only the iron nail

20. A spring balance shows 3 N for an object. Its mass is about (taking 10 N for 1 kg): 1

(a) 30 kg **(b)** 0.3 kg **(c)** 3 kg **(d)** 0.03 kg

21. A tornado forms mostly: 1

(a) at the poles **(b)** over the deep sea always
(c) over land under thunderclouds **(d)** in caves

22. Rubbing a drop of spirit on the palm gives a cooling feeling because spirit: 1

(a) is cold **(b)** burns
(c) is heavy **(d)** evaporates quickly

23. Electric wires are made of copper because copper is: 1

(a) magnetic **(b)** brittle
(c) light and non-conducting **(d)** a good conductor and ductile

24. In a fair test on the effect of stirring, which must be kept the same? 1

(a) stirring **(b)** temperature and amount of sugar
(c) colour **(d)** time to dissolve

25. A concave lens forms an image of an object. The image is: 1

(a) virtual, erect and smaller **(b)** virtual and larger
(c) real and larger **(d)** real and inverted

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26. The full Moon rises at about: 1

(a) midnight **(b)** sunrise **(c)** noon **(d)** sunset

27. A tiger population in a forest falls. The deer population will probably: 1

(a) decrease **(b)** increase **(c)** remain zero **(d)** vanish

28. In 200 L of air the volume of nitrogen is about: 1

(a) 42 L **(b)** 78 L **(c)** 156 L **(d)** 21 L

29. A student gets times of 12.1 s, 11.9 s, 12.0 s for 20 swings. The mean is: 1

(a) 12.1 s **(b)** 11.9 s **(c)** 36.0 s **(d)** 12.0 s

30. A student sees rectangular cells with a clear nucleus but no green bodies. The cells may be from: 1

(a) a human cheek **(b)** an onion bulb **(c)** bacteria **(d)** a leaf

31. Mosquito larvae are found in water that is: 1

(a) flowing fast **(b)** frozen **(c)** stagnant **(d)** boiling

32. A compass needle placed near a battery wire does not deflect. A likely reason is: 1

(a) the switch is open **(b)** the needle is a magnet

(c) the wire is copper **(d)** the cell is new

33. Two persons pull a rope with 100 N each in opposite directions. The net force on the rope is: 1

(a) 200 N **(b)** 50 N **(c)** 0 N **(d)** 100 N

34. A brick of weight 20 N has faces of areas 0.1 square metre and 0.04 square metre. The minimum pressure it can exert when resting on a face is: 1

(a) 500 Pa **(b)** 200 Pa **(c)** 0.2 Pa **(d)** 2 Pa

35. Which of the following sets are in the right order of increasing space between particles? 1

(a) gas, liquid, solid **(b)** liquid, solid, gas

(c) solid, liquid, gas **(d)** gas, solid, liquid

36. A student heated a mixture of iron and sulphur strongly and got a black solid that is not magnetic. The black solid is a: 1

(a) element **(b)** mixture **(c)** alloy **(d)** compound

37. How much water is needed to make 500 g of a 5 per cent salt solution? 1

(a) 475 g **(b)** 450 g **(c)** 25 g **(d)** 500 g

38. A man walks towards a plane mirror at 2 m/s. The speed of his image towards him is: 1

(a) 0 **(b)** 2 m/s **(c)** 1 m/s **(d)** 4 m/s

39. If today is a full moon, the next new moon will be after about: 1

(a) 7 days **(b)** 15 days **(c)** 30 days **(d)** 60 days

40. In a food chain, the amount of energy available is greatest at the: 1

(a) producer level **(b)** secondary consumer level

(c) primary consumer level **(d)** top carnivore level

41. The Earth's water is about 97 per cent salt water. If total water is 1000 units, the units of fresh water are about: 1

(a) 300 **(b)** 970 **(c)** 3 **(d)** 30

42. Which of the following is a controlled variable when comparing the cooking of puris at 150 and 180 degrees C? 1

(a) colour at the end **(b)** thickness of dough
(c) puffing time **(d)** oil temperature

43. Which of the following is the correct statement about plant cell vacuoles? 1

(a) They are absent
(b) They are large and store water and nutrients
(c) They make food
(d) They are the control centre

44. A health camp checks weight, height and haemoglobin. This is: 1

(a) vaccination **(b)** mental health therapy
(c) monitoring physical health **(d)** treatment of an infection

45. A student connects two cells in series in a circuit with a bulb. The bulb: 1

(a) glows dimmer **(b)** goes out **(c)** glows brighter **(d)** does not glow

46. An electroscope's leaves collapse when a charged rod is touched by a finger because: 1

(a) air enters **(b)** charge flows to earth
(c) leaves become heavy **(d)** a magnet acts

47. The sea breeze occurs because: 1

(a) sea is frozen **(b)** sea is warmer at noon
(c) land is cooler than sea in the daytime **(d)** land is warmer than sea in the daytime

48. Water at 4 degrees Celsius and 90 degrees Celsius: the faster-moving particles are in: 1

(a) water at 90 degrees Celsius **(b)** water at 4 degrees Celsius
(c) equal in both **(d)** cannot be said

49. Which element is both lustrous and brittle? 1

(a) iron **(b)** copper **(c)** gold **(d)** silicon

50. Which of these settles on standing? 1

(a) milk **(b)** sand in water **(c)** sugar in water **(d)** salt in water

END OF QUESTION PAPER **Total: 50 Marks**

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